Supply Chains Under Stress

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In October 2021 the Maersk Emden waited sixteen days off Long Beach with a Mississippi firm’s Christmas toys on board.

YOUR IB VOICE

FROM TOYOTA TO LONG BEACH

Goodman traces the shortages of 2021 back to a factory manager in postwar Japan. Using one example from the chapters, explain how a practice that made firms more efficient also made them more exposed to a shock.

10 minutes · Individual · Notes permitted

Why did the world run out of everything in 2021?

Factories in China shut down, so there was less to ship.

People stuck at home bought more goods, and firms ordered extra to be safe.

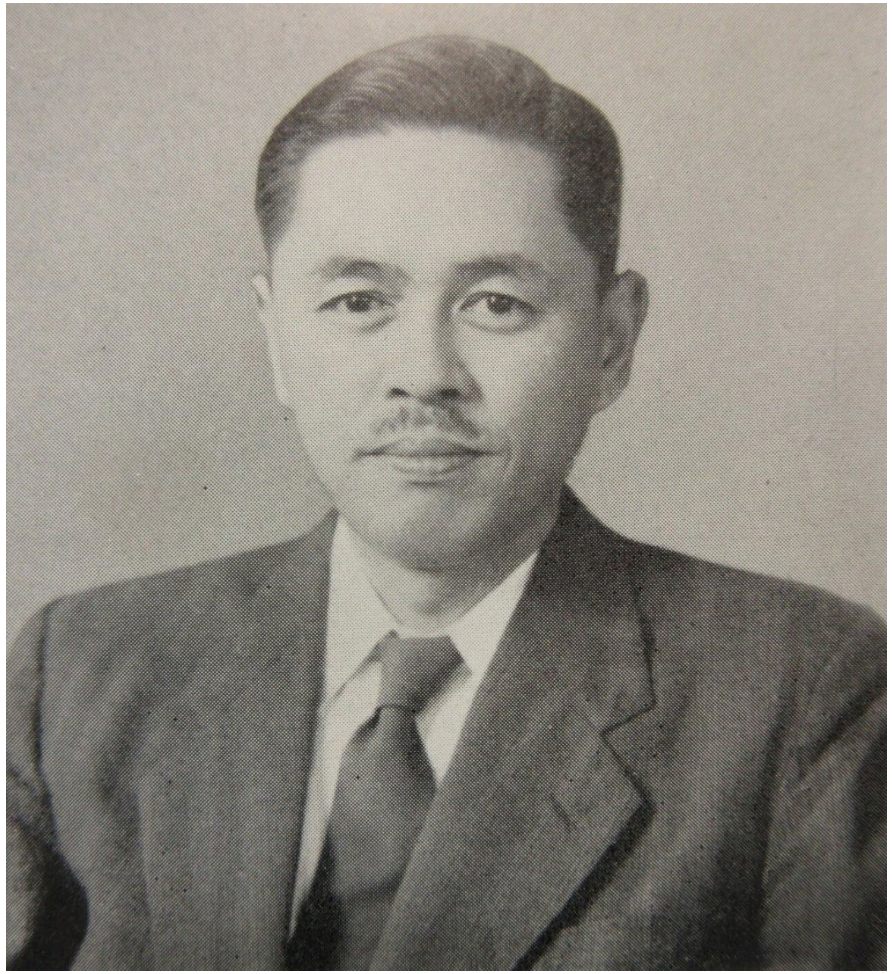
Firms had kept so little stock that any delay became a shortage.

The companies that ran the ports gained from the delays and were slow to end them.

> Which of these would have caused shortages even without a pandemic?

Why did firms keep so little stock?

Taiichi Ohno made only what the next step used



Toyota Motor Co. history committee, Taiichi Ohno, 1958 · Wikimedia Commons, public domain

In 1950 Toyota built more trucks than it could sell and nearly went bankrupt.

Ohno copied the supermarket, which restocks only what customers take.

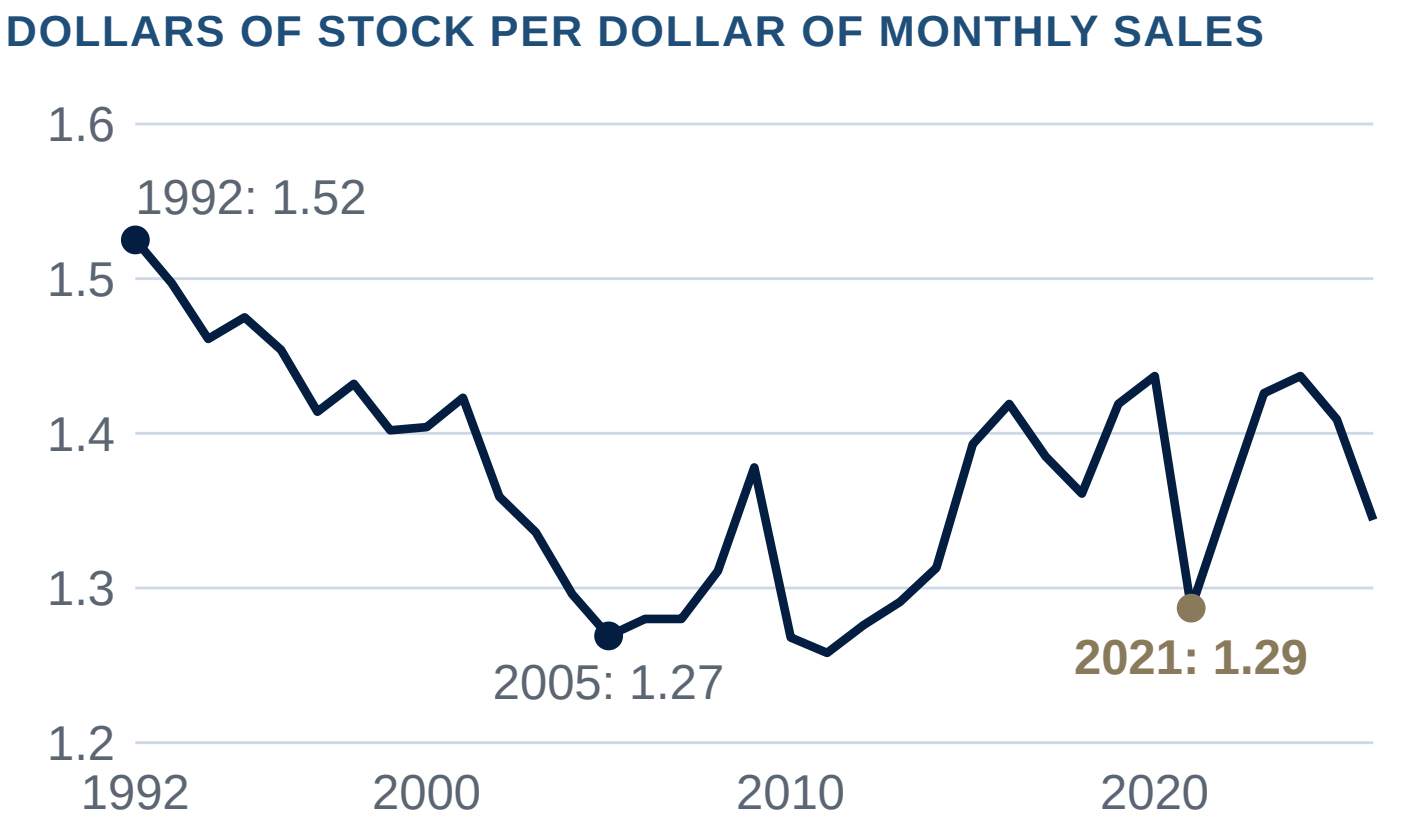
> *Just in Time* means making and delivering each part only when the next step needs it.

U.S. firms learned to hold less stock

By the late 1980s two-fifths of the largest U.S. manufacturers used some version of Just in Time.

Lean stock freed cash in normal years and left little to draw on in a bad one.

> In 2021 firms ran their shelves down and could not refill them.



U.S. Census Bureau, Total Business: Inventories to Sales Ratio, via FRED, 2026: annual averages.

Ohno wrote that his ancestors stored rice for disasters, and that business must give up such hoarding. Was he wrong, or did his followers misread him?

> What part of Ohno’s system did the consultants leave out?

Why could nobody clear the ports?

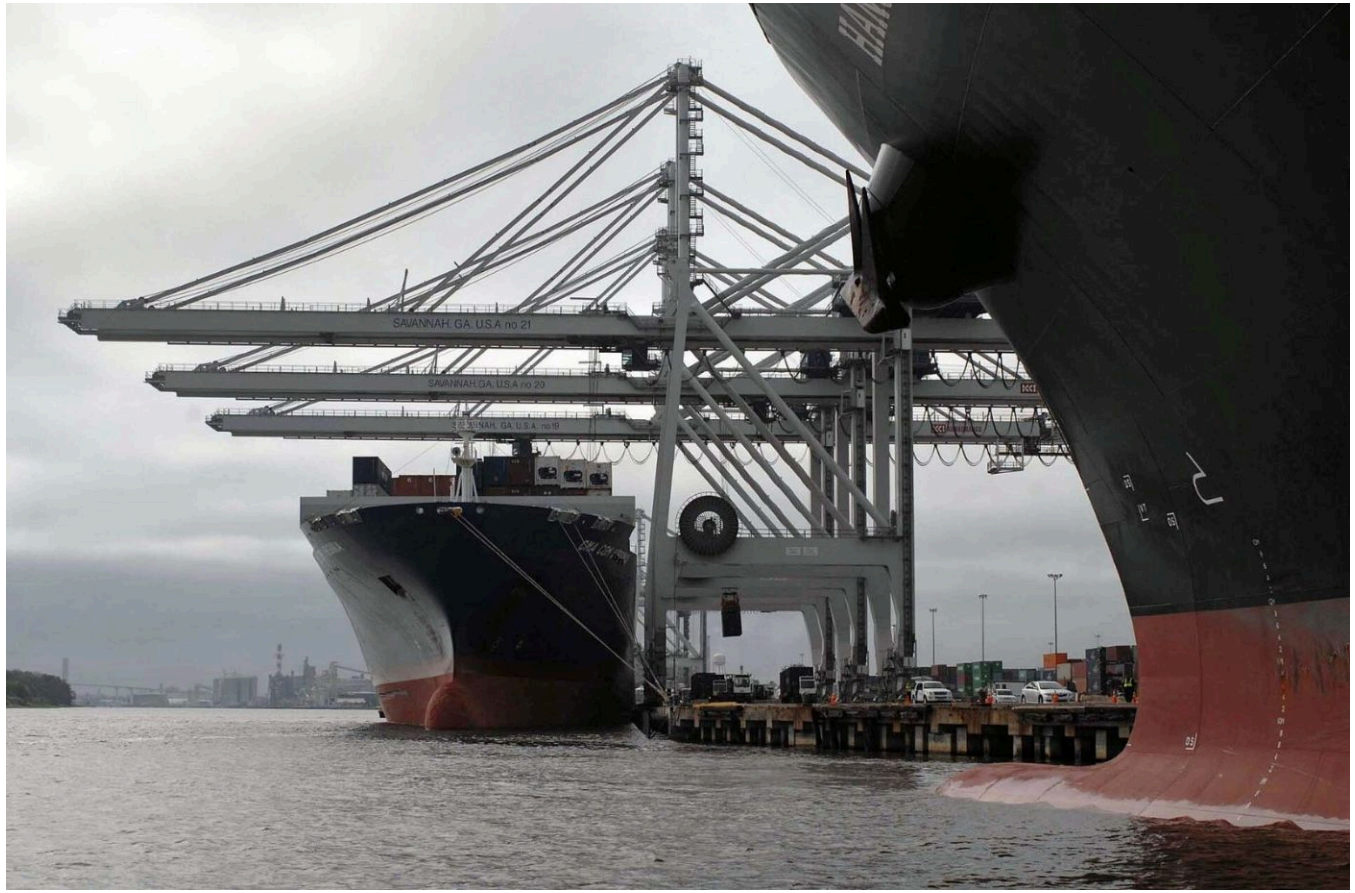
Dozens of ships waited off Los Angeles



European Union, Copernicus Sentinel-2 imagery, October 10, 2021 · Wikimedia Commons, attribution

A *bottleneck* is the step with the least spare capacity, which sets the pace of the whole chain.

The jam moved inland, past the berth



U.S. Army Corps of Engineers, Savannah Harbor, Garden City Ocean Terminal · Wikimedia Commons, public domain

Warehouses near the ports were full, so containers sat on their truck chassis instead of being unloaded.

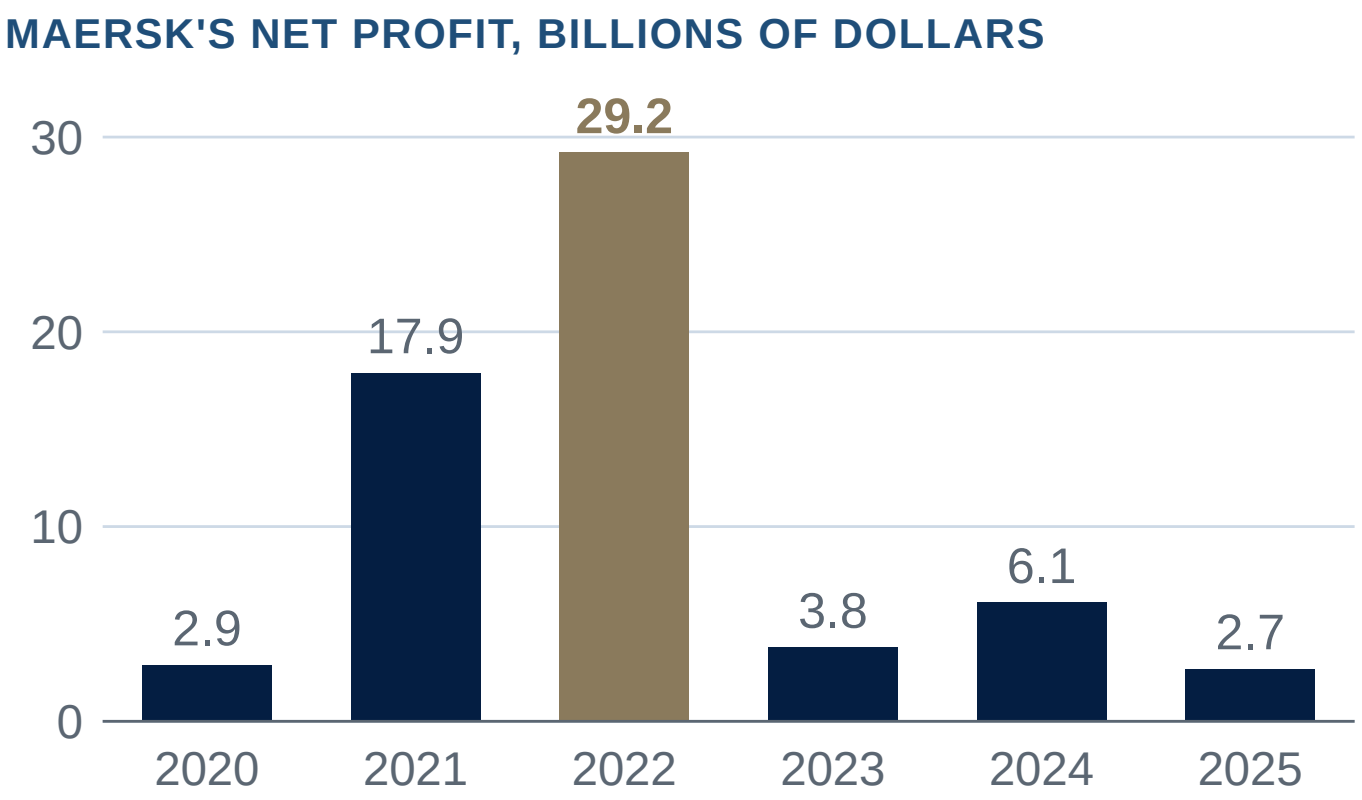
Without free chassis, trucks could not collect the next boxes, and the yards filled.

> “If there’s no space out here, it doesn’t matter if I have fifty berths.”

In October 2021 President Biden had the ports of Los Angeles and Long Beach promise to work around the clock. Why did that not end the jam?

> Who could have added capacity, and who gained from not adding it?

Maersk earned record profits during the jam



A.P. Moller-Maersk, annual results, 2020 to 2025: net profit.

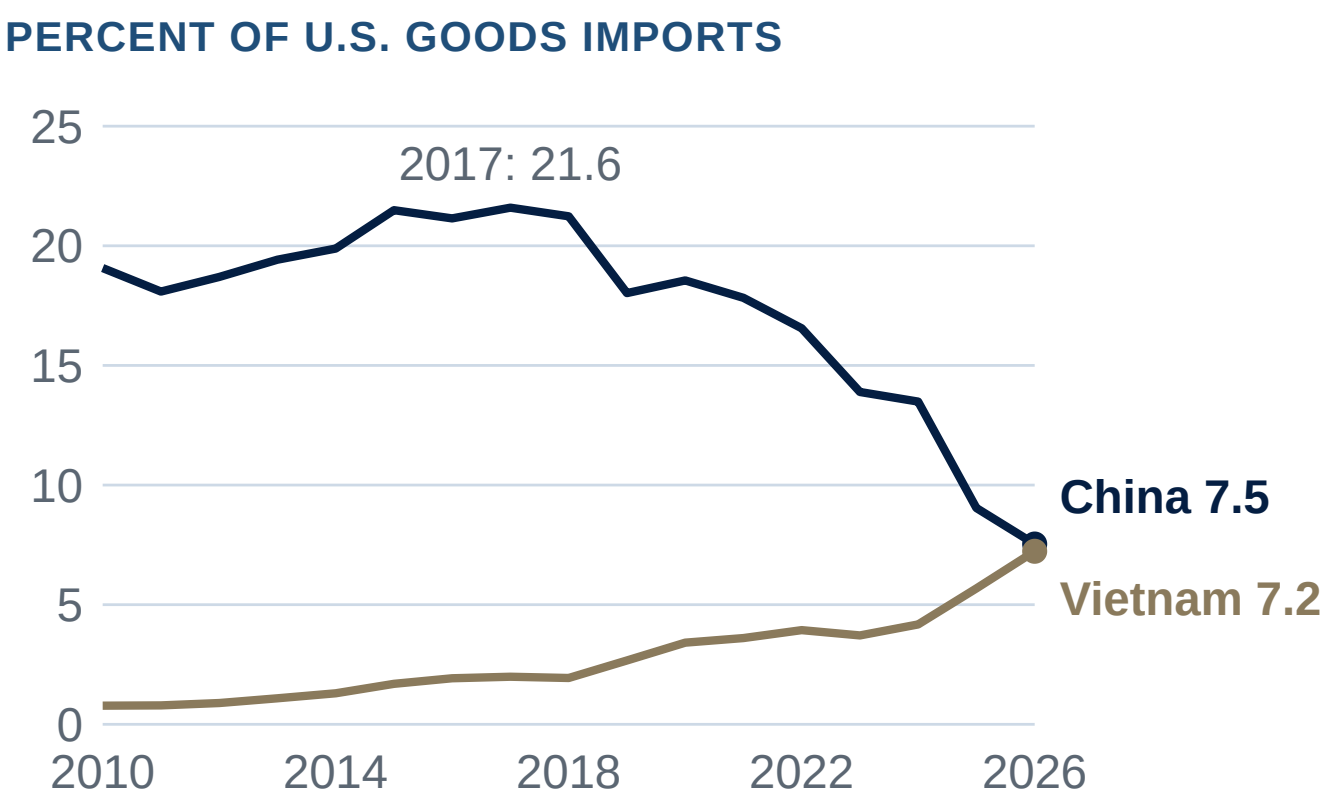
Maersk ran one of the largest terminals in Los Angeles as well as the ships waiting outside.

Its chief executive told investors it would not sell all its capacity unless demand was there.

> A firm that controls a bottleneck can profit from it.

Can a firm buy resilience?

U.S. buyers shifted orders from China to Vietnam



U.S. Census Bureau, imports of goods by country, via FRED, 2026: China, Vietnam, and all countries.

In 2019 an advisor urged Glo to spread its orders across factories in more than one country.

In 2026 the United States is buying almost as much from Vietnam as from China.

> *Supply-chain resilience* is the ability to keep supplying when one source or route fails.



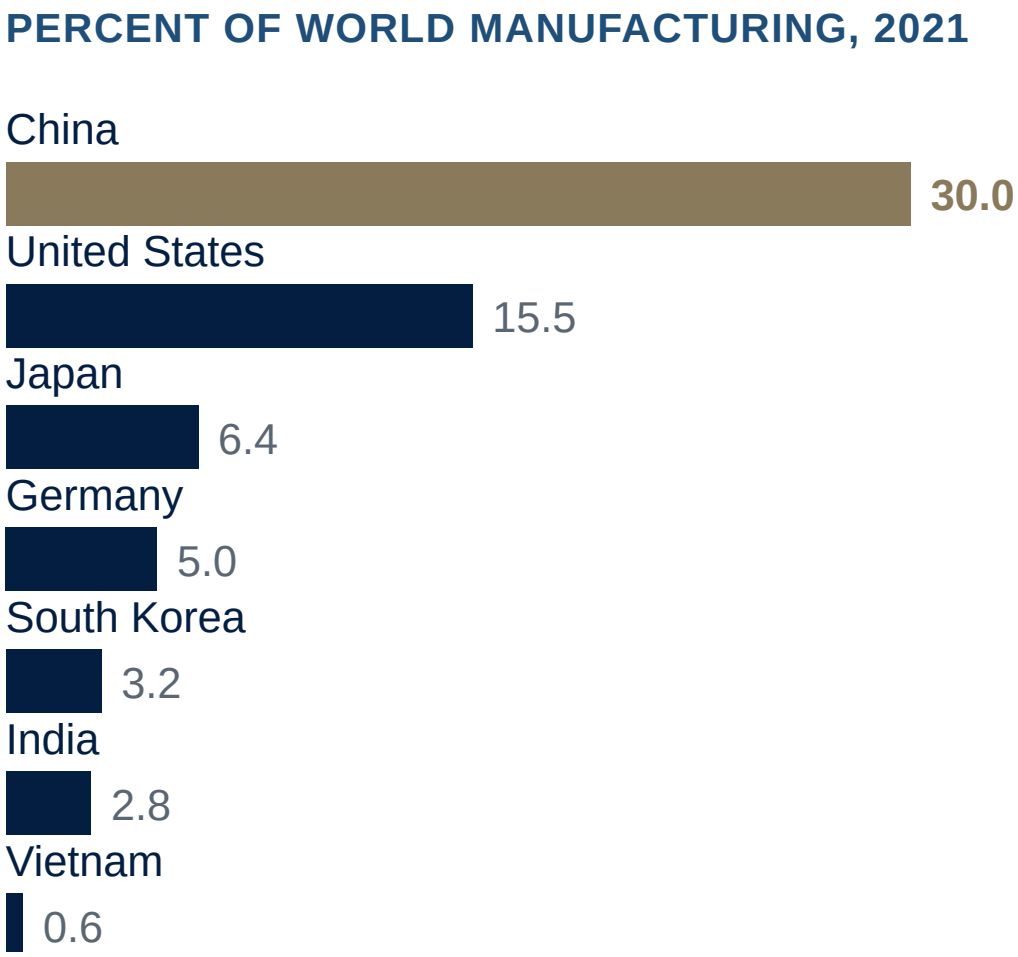
“We’re not even breaking up with China. We’re just asking if we can also see other people at the same time.” Barker, of Glo

China makes nearly a third of the world's goods

Walker posted a request for a display screen on Alibaba and had twenty quotes from Chinese factories within minutes.

Factories in Vietnam often import their parts from China.

> Goodman calls this a *barrier to exit*: the scale that makes China cheap also makes leaving it hard.



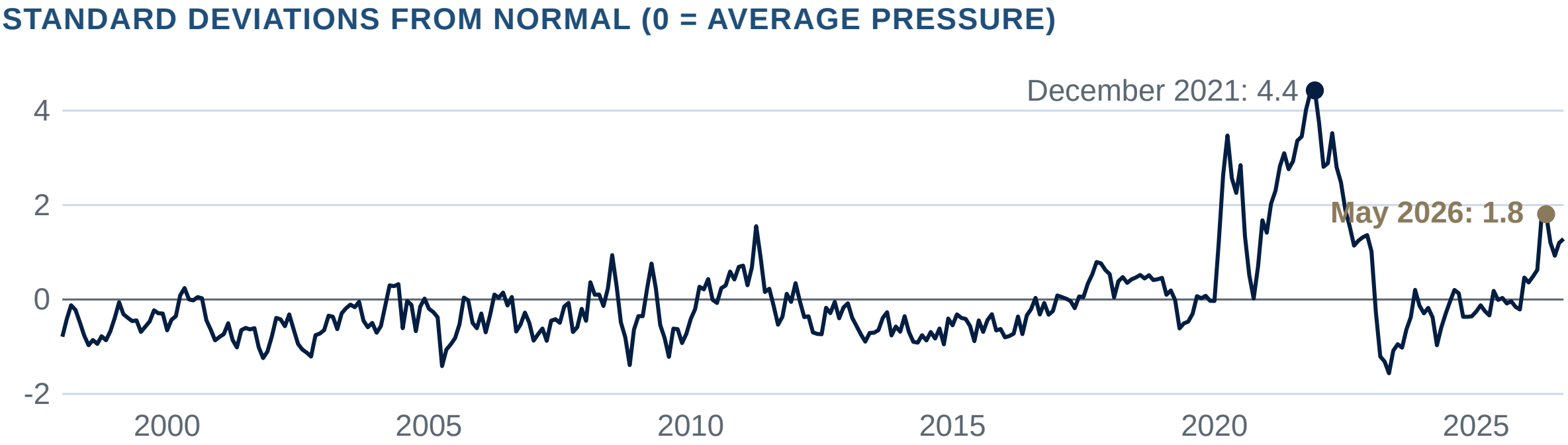
World Bank, World Development Indicators, 2026: manufacturing value added, 2021.

Goodman warns that moving production from China to several countries could leave firms more exposed to a shock, not less. How can that be?

> When is a second supplier not really a second supplier?

What has closing Hormuz done to prices?

Supply-chain pressure rose again in 2026



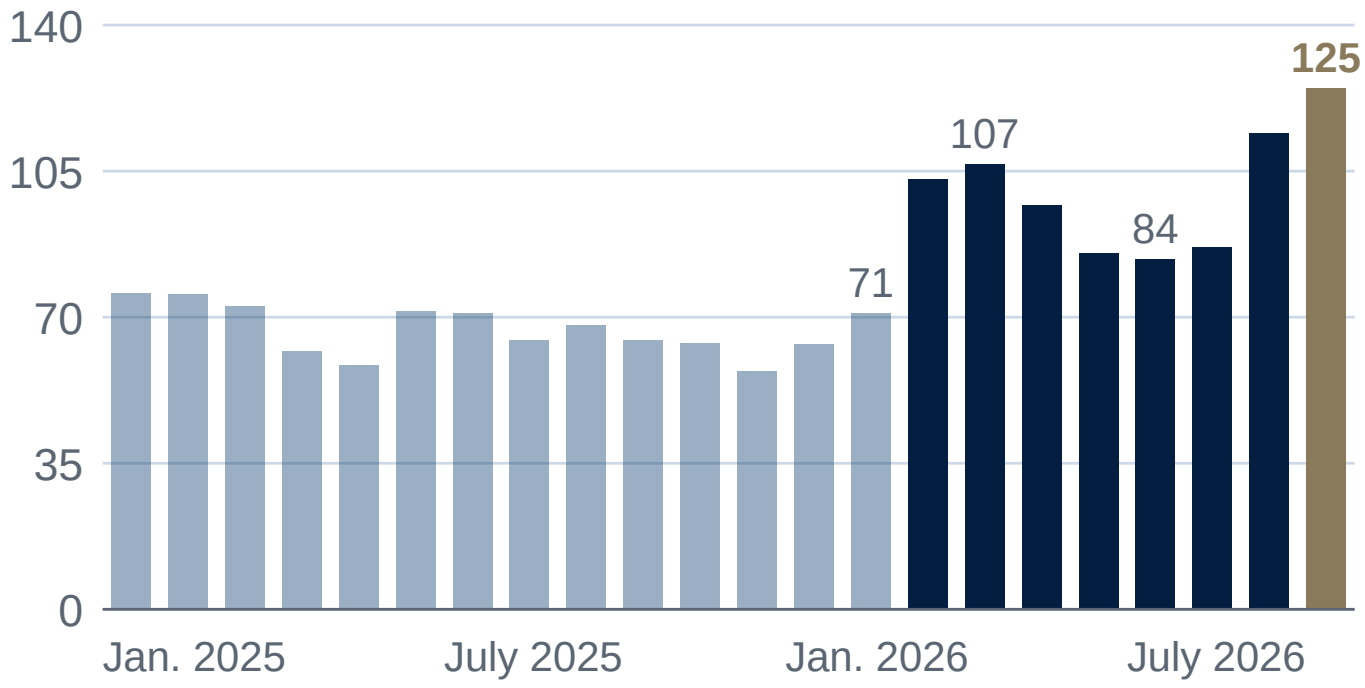
Federal Reserve Bank of New York, Global Supply Chain Pressure Index, 2026.

The war with Iran has closed most shipping through the Strait of Hormuz since March.

> A chain built only for normal years will break again.

Brent averaged $125 a barrel in October

BRENT CRUDE, DOLLARS PER BARREL



U.S. Energy Information Administration, Europe Brent spot price, via FRED, 2026: monthly averages.

Traffic through Hormuz is still a fraction of normal, and the U.S. still blockades Iran's ports.

On October 30 Iran said four tankers had struck mines in the strait.

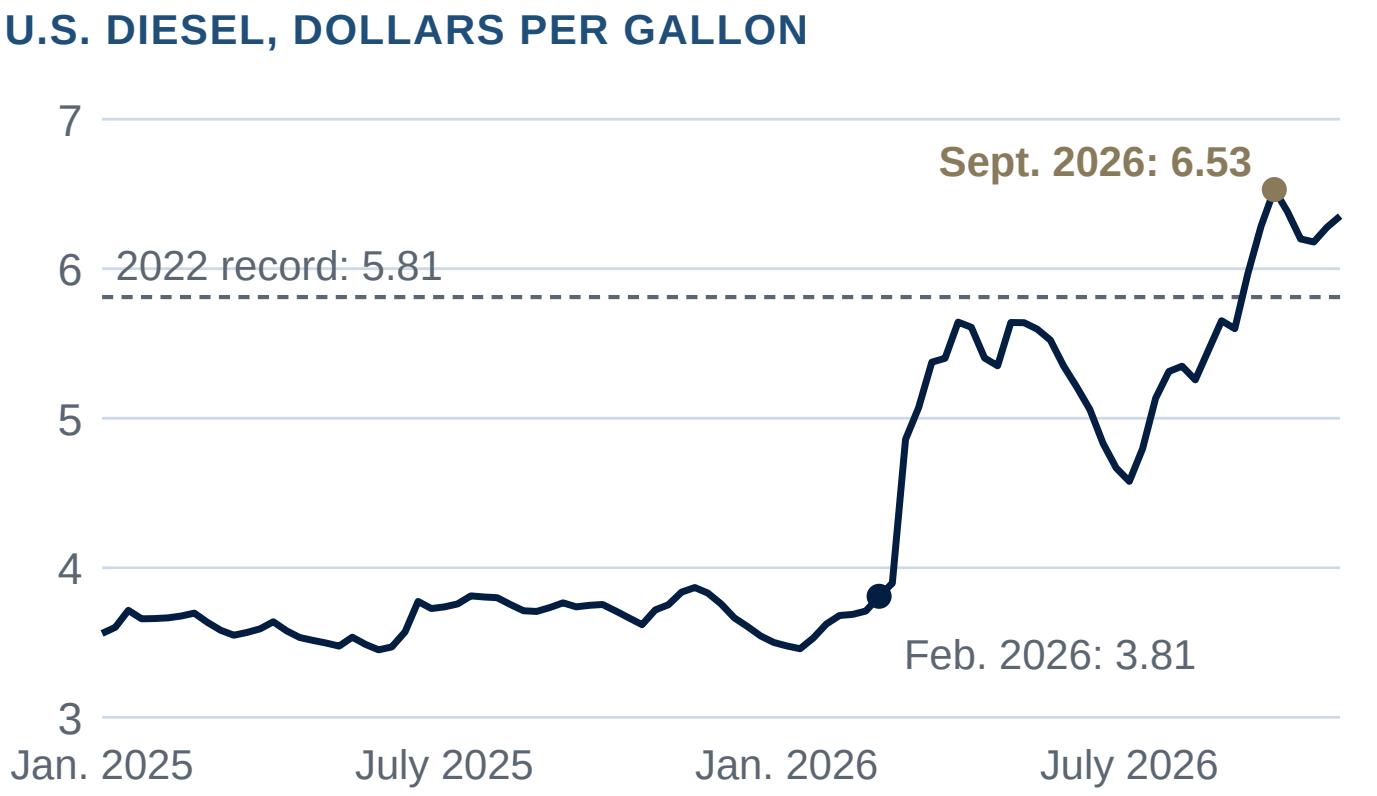
> Oil has one world price, so a closed strait raises what every buyer pays.

U.S. diesel set a record in September

Diesel moves trucks, trains, ships, and farm machinery, so its price reaches almost every good.

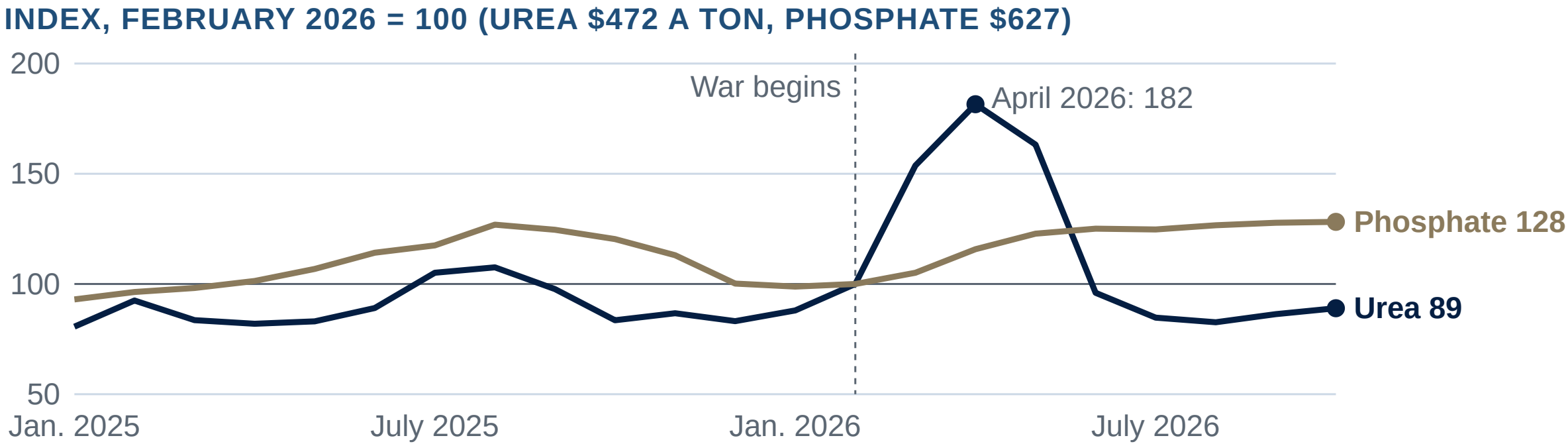
It rose faster than crude because Gulf refineries that supplied Europe stopped shipping.

> A shock to one input raises the cost of moving everything else.



U.S. Energy Information Administration, diesel retail price, via FRED, 2026: weekly.

Urea fell back, but phosphate kept rising



World Bank, Commodity Price Data (Pink Sheet), November 2026: urea, Middle East, and diammonium phosphate.

Urea's price fell once China resumed exports. Phosphate fertilizer needs Gulf sulfur and ammonia, which stayed scarce.

> If farmers use less fertilizer, harvests shrink, and the shock reaches food a season later.

Plan Glo’s next Christmas

1. **Trace** Glo’s light-up cube from the factory in Ningbo to a store shelf in the United States, listing each step from the chapters.
2. **Diagnose** the step most likely to stop the whole chain, using one passage from Goodman.
3. **Choose** one buffer for that step: extra stock in Mississippi, a second factory outside China, an earlier order date, or air freight for part of the order.
4. **Defend** your choice to the person next to you, including what it costs in a normal year.

15 minutes · Individual, then pairs · Readings permitted

Which step did you choose as the bottleneck, and what did your buffer cost in a normal year?

> Which of the buffers would also protect Glo if trade with China were cut off?

Ohno said business needed the courage to buy only what is needed when it is needed. After 2021, how much extra should a firm hold, and who should pay for it?

> What would a buffer look like for a firm whose chain runs through software?

How do shocks travel through supply chains?

Just in Time cut stock and spare capacity, so a delay anywhere soon became a shortage everywhere.

A chain moves only as fast as its narrowest step, and in 2021 that step moved from factories to ports to warehouses.

A firm that controls a bottleneck can profit from it and may be slow to clear it.

Resilience costs money in normal years, and many alternatives to China still depend on China.

A strait closed in the Gulf raises the cost of fuel and fertilizer for every chain that uses them.